

Ocean West

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AI ENGINEERING PROJECTS

- CLU: Multi-Agent AI Assistant | Python, Claude/GPT/Ollama, FastAPI** 2026 - Present
- Architected a production multi-agent system: **68 tools** across **16 specialized sub-agents** with a provider-agnostic backend supporting Claude, GPT, and local Ollama models; **418 passing tests** across about 28,000 lines of Python.
 - Designed tiered autonomy with human-in-the-loop approval gates enforced at the tool registry: read-only tools run unattended, while any action that writes files or incurs cost requires explicit approval.
 - Built an LLM evaluation framework scoring output on citation resolution, grounding, and task completion; caught a fabricated-source failure (35% vs 96% on verified output) that passed all 418 unit tests.
- Composites AI Assistant (Longhorn Racing) | Python, Streamlit, ChromaDB, Ollama** Spring 2026
- Built a local-first RAG system over **200+ composites documents** (textbooks, material standards, ANSYS documentation, internal process files) with on-device embeddings and a local Chroma vector store, replacing manual shared-drive searches.
 - Adopted as the standing reference tool by the **10-person composites subsystem**; implemented a strict-local privacy mode blocking any cloud backend at the application layer to keep proprietary engineering documents on-machine; extends an Excel-based laminate calculator that cut design iteration time 15%.
- Sales Utility Engine | FastAPI, Pydantic, SQLite, Ollama** Summer 2026
- Led development of an end-to-end automotive sales assistant with separate customer and employee workspaces: inbound lead intake, chat-based qualification, vehicle catalog recommendations, meeting booking, and sales-handoff summaries.
 - Built a rule-based qualification engine scoring profile completeness, sales priority, and booking readiness over SQLite persistence and Pydantic models, with local Gemma inference through Ollama, deterministic fallback, and 23 passing tests.
- Door Guard | Python, YOLO, OpenCV, Ollama** Summer 2026
- Developed a local-first computer-vision prototype detecting people in live webcam and RTSP streams with configurable door-zone loitering tracking, risk-based snapshots, JSONL event logs, and local LLM-generated deterrent dialogue with text-to-speech.

TECHNICAL SKILLS

AI Engineering: Python, Retrieval-Augmented Generation, OpenAI and Azure OpenAI APIs, ingestion and embedding pipelines, ChromaDB, local-first inference, multi-agent orchestration, agent evals and observability, semantic and BM25 retrieval, structured outputs

Software and AI Operations: FastAPI, Streamlit, REST APIs, CLI development, Pydantic, JSON Schema, OpenAPI, Pytest, Git/GitHub, API health and metrics, JSONL audit logging, LLM guardrails, prompt-injection testing

Computer Vision and Engineering: YOLO, OpenCV, RTSP/IP camera streams, SolidWorks, ANSYS Mechanical, ANSYS ACP, ASME/ISO/ASTM standards, additive manufacturing (FDM), tooling design, MATLAB, Arduino, data acquisition

ENGINEERING EXPERIENCE

- AI Engineering Intern** Summer 2026
Agentic Innovations *Texas, United States*
- Completed a 12-week AI engineering program across engineering, business, and marketing tracks; delivered prototypes on a 4-person cross-functional team (2 engineering, 1 business, 1 marketing).
 - Led Phase 2 as majority contributor on SUE; presented it to a panel of senior engineers from NVIDIA, Google, Meta, and NASA alongside cybersecurity specialists and industry recruiters.
- Composites Materials and Processes Integration Lead, Combustion Team** May 2025 - May 2026
Longhorn Racing, UT Formula SAE *Austin, TX*
- Led the materials and processing subsystem for non-aerodynamic composites on a 90-person Formula SAE team, owning composite design decisions across five vehicle systems; built material-property workflows used across the 10-person subsystem and referenced by other systems verifying their own calculations.
 - Supported a steering-wheel redesign achieving a **50% mass reduction**; validated ANSYS simulations against tensile, bend, adhesive, and torsional testing to establish factors of safety from 1.5 to 8, and authored testing methodology adopted for the following season.
- Undergraduate Research and Development Intern** Jan 2024 - Aug 2024
UTSA Klesse College of Engineering and Integrated Design *San Antonio, TX*
- Conducted **100+ axial fatigue tests** on additively manufactured Ti-6Al-4V specimens for an **Air Force Research Laboratory** project, including Equivalent Initial Damage Size analysis; results supported a graduate researcher's findings presentation.

EDUCATION

- The University of Texas at Austin** Expected Dec 2026
B.S. Mechanical Engineering, GPA: 3.51/4.00 *Austin, TX*
- The University of Texas at San Antonio** Aug 2022 - May 2024
Mechanical Engineering Studies, GPA: 3.89/4.00 *San Antonio, TX*

ADDITIONAL

Awards: Formula SAE: 26th Overall 2026 (from 46th, best in 4 years), 11th in Design 2025 | President's List | Dean's List (2x)
Interests: Motorsport and Formula 1, golf, astronomy, photography